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Effects of *Echinacea purpurea* supplementation on markers of immunity, health, intake, and growth of dairy calves

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ABSTRACT

Echinacea purpurea (EP) is an herb that has demonstrated immunostimulatory and anti-inflammatory effects with the potential to improve immunity, health, and performance in animals. The objective of this study was to investigate how supplementing calves with EP affects their blood immunity marker profile, health, intake, and growth. Male Holstein calves (n = 240), sourced from local dairy farms or auction, arrived at a rearing facility between 5 and 14 d of age and were kept in individual pens in 1 of 3 rooms (80/room) for 56 d, and then put into groups for the remaining 21 d of the trial. Calves received milk replacer (MR) 2× per day for 56 d (total = 36 kg of MR) and had ab libitum water and starter access. Within room, calves were randomly assigned to 1 of 3 treatments: (1) control (n = 80), (2) 3g of dried (powder) EP extract per day split over 2 milk feedings from experiment d 14–28 (n = 80), and (3) 3 g of dried (powder) EP extract per day split over 2 milk feedings from experiment d 1–56 (E56; n = 80). The powdered EP treatments were mixed into the liquid MR. On d 1, 14, 28, and 57 rectal temperatures and blood were collected from a subset of calves (n = 117; 39 calves/treatment), and blood serum was assessed for serum total protein (d 1), haptoglobin, white blood cells, and cytokines. Failed transfer of passive immunity was defined as serum total protein <5.2 g/dL. Calves were health scored 2× per day, receiving fecal and respiratory scores until d 28 and 77, respectively. Calves were weighed on arrival and then weekly until d 77. Milk replacer and feed refusals were recorded. Supplementation of EP was associated with lower haptoglobin levels, segmented neutrophil counts, segmented neutrophil per lymphocyte ratio, respiratory scores in auction derived calves, and higher lymphocyte counts and d 28 rectal temperature. Of calves with heavier arrival body weight, E56 calves had greater postweaning weekly body weight. There was no detected effect of EP supplementation on total white blood cells, band neutrophil, monocyte, and basophil counts, IL-10, IL-6, and TNF-α levels, fecal scores, risk of receiving diarrhea and respiratory treatment, risk of bovine respiratory disease (calves were deemed at risk for bovine respiratory disease if they had at least 1 respiratory score ≥5), risk of mortality, MR and feed intake, average daily gain, and feed conversion ratio. Overall, EP supplementation to dairy calves was associated with immunomodulation and reduced inflammation, evidenced through blood markers, although only few minor health and growth improvements were observed. Benefits were observed particularly when fed across the whole milk feeding period.

Key words: feed additive, medicinal herb, phytotherapy

INTRODUCTION

Statistics on preweaning dairy calf morbidity and mortality remain undesirable. A recent estimate of mortality rates of female dairy calves in Canada was 6.4% (n = 1,373 farms; Winder et al., 2018), while it was similarly 6.9% in the United States considering female and male dairy calves (n = 105 farms; Walker et al., 2012). Several researchers have documented high levels of morbidity, with 33.9% of female dairy calves being treated for disease in Urie et al. (2018; n = 2,545 calves) and 23 and 22% of calves were treated for diarrhea and respiratory disease, respectively, in Windeyer et al. (2014; n = 2,874). Male dairy calves also have a high level of mortality and morbidity, with 7.5% dying and 88.4% being treated for a health disorder at a veal facility (n = 992 calves; Goetz et al., 2021). These high levels of disease result in high antimicrobial use. Walker et al. (2012) reported 73 and 82% of calves that experienced diarrhea and respiratory disease received an antimicrobial, respectively. Likewise, Urie et al. (2018) reported approximately three-quarters (73.8%) of sick calves were administered an antimicrobial. High antimicrobial use is a concern for many reasons – especially the growing challenge of antimicrobial resistance (Berge

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et al., 2006). To address these levels of morbidity and mortality, evaluating colostrum management, housing design and hygiene, nutrition, pathogen exposure, and stressors including transportation (Hulbert and Moisá, 2016; Seckin et al., 2018) can aid in the prevention of disease.

Just as in human medicine, natural supplements may be a useful intervention to support and enhance dairy calf immunity and health, potentially serving as a complementary treatment option or alternative to antimicrobials (Seckin et al., 2018). Echinacea is an herbaceous perennial plant indigenous to North America (Barnes et al., 2005), commonly called purple coneflower (Kumar and Ramaiah, 2011). There are 9 species of Echinacea, although *Echinacea purpurea* (**EP**), *Echinacea angustifolia*, and *Echinacea pallida* are the 3 used medicinally, with EP being the most common (Barnes et al., 2005). Active components in EP include: (1) phenolics which exhibit antioxidant effects (Dalby-Brown et al., 2005), (2) alkamides which have anti-inflammatory effects (Chen et al., 2005), and (3) polysaccharides which have immune stimulating abilities (Steinmüller et al., 1993). There is also more recent research on bacterial endophytes in EP and their ability to enhance the immune system, accounting for the majority of the immunomodulation observed (Pugh et al., 2013; Todd et al., 2015; Haron et al., 2016). Specifically, lipopolysaccharides and Braun-type lipoproteins derived from the bacterial endophytes in EP stimulate the body's macrophages (Tamta et al., 2008; Pugh et al., 2013), which promotes the production of cytokines (both proinflammatory, such as TNF-α and IL-6, and anti-inflammatory, such as IL-10; Burger et al., 1997; Sullivan et al., 2008). The effect of bacterial endophytes is supported by research that demonstrated EP grown in a sterile environment lacked both lipopolysaccharides and macrophage stimulating ability (Todd et al., 2015). Furthermore, the addition of substances that degrade bacterial lipopolysaccharides and lipoproteins (polymyxin B and lipoprotein lipase) resulted in diminishment of the immunomodulation activity (Tamta et al., 2008). *Echinacea purpurea* is also reported to increase circulating white blood cells (Goel et al., 2005). Therefore, EP can affect the innate immune system (Haron et al., 2016). Considering adaptive immunity, immunoglobulins have also been investigated although there are conflicting results: IgG and IgM levels were reported to increase (Rehman et al., 1999) and decrease (Mishima et al., 2004) in response to *Echinacea*.

Even though *Echinacea* has been consumed by humans for medicinal reasons for centuries (Brush et al., 2006), its effects on health and immunity are still not completely understood (Seckin et al., 2018). Furthermore, there is a lack of research on the effects of medicinal plants on young livestock species (Ayrle et al., 2016). The effects of EP supplementation on dairy calves remains under-researched to date and has been largely investigated in combination with other supplements, yielding variable results. Specifically, Ismael et al. (2009) reported that co-administering *Echinacea* with foot and mouth disease vaccines increased the vaccine's efficiency in calves. Seckin et al. (2018) reported that supplementation of EP and *Pelargonium sidoides* to calves elicited an immune response, evidenced through increased IgG and γ-IFN levels and upregulation of γ-I IL-1-β, IL-2, and TNF-α. Finally, Ayrle et al. (2021) suggested that EP may activate the local enteric immune system, however, they detected mixed results with EP supplementation in calves. Therefore, further investigation with supplementation of EP alone is warranted.

The objective of this study was to determine the effects of EP supplementation on markers of immunity, health, feed intake, and growth of dairy calves. The hypothesis was that EP supplementation, especially to calves during the whole milk-fed period, would reduce inflammation (i.e., lower haptoglobin concentration) and stimulate the immune system (i.e., increased cytokine and white blood cell counts), and consequently, improve the health, intake, and growth of calves compared with calves not receiving EP.

MATERIALS AND METHODS

Animals and Housing

A randomized clinical trial was conducted at a commercial calf rearing facility (Mapleview Agri Ltd., Palmerston, Ontario, Canada), using 3 batches of 80 calves, for a total sample size of 240 male Holstein calves. For each batch, all calves arrived at the facility on 1 d (November 9, 30, and December 21, 2020), sourced from auction barns (n = 103 calves) or directly via 2 drovers (drover 1 = 53 calves, and drover 2 = 84 calves), from approximately 30 local Ontario dairy farms. Each group of 80 calves were housed in structurally identical, but different rooms at the facility. All calves that arrived were enrolled (no calves were excluded at enrollment). All calves had an unknown history, each with an age estimated to be between 5 to 14 d of life. For each batch, upon arrival, all calves were housed in 1 room (Supplemental Figure S1; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023) that had 4 rows of 20 metal bar-sided stalls (80 stalls total) with a walkway between each row of stalls. Each calf was randomly assigned to a stall (101.60 × 78.74 × 121.92 cm; height × width × length) with rubber slatted floors and no bedding (as per facility

standard) for the milk-fed period (d 0 to 56). Each stall had an opening on the front, which allowed the calves to reach through and consume their milk or solid feed out of a bucket and trough, respectively. The buckets were attached to a metal bar that allowed them to be rotated into position for the calves at milk feeding times, and then rotated beneath the feed trough, out of the way, after milk feeding times. Calves had auditory and visual contact with other calves, however, physical contact was limited due to the metal bar stall sides. For the postweaning period (d 57 to 77), the gates separating each set of 5 consecutive stalls were removed, which consequently incorporated the previous walkway into the housing area to allow calves in those stalls to co-mingle and have physical contact with each other in a larger space; each group of 5 calves was referred to as a pod (101.60 × 198.12 × 396.24 cm; height × width × length). The ventilation system in the rooms consisted of an air inlet on one side and 5 chimney fans to exhaust the air. All study procedures were reviewed and approved by the University of Guelph Animal Care Committee (AUP#4134).

Feeding and Health Management

All calves were offered a skim-milk based milk replacer (**MR**; 26% CP, 17% fat; Mapleview Agri Ltd.; Table 1). The fat sources in the MR were lard, palm, and coconut. The MR contained no feed additives, including no antimicrobials. Barn staff individually fed the calves MR according to the feeding schedule outlined in Table 2, twice a day (at 0600 and 1630 h) by bucket with a floating nipple to drink from. During the first milk feeding, barn staff trained the calves to consume milk out of a bucket via the floating nipple by allowing the calves to suckle on their fingers and guiding their mouth down onto the nipple. Staff supervised the calves for the first 14 d to try to get as many calves as possible to use the nipple, although some calves preferred to drink straight from the bucket resulting in the nipples being used approximately 90% of the time. All milk rejections were recorded. All calves were completely weaned and received no milk on d 57. For the milk-fed period, calves had ad libitum access to water provided through a nipple water dispenser in each pen, whereas for the postweaning period, each pod had ad libitum access to water provided through a water bowl. Ad libitum solid feed was offered from d 1 in a trough, fed per pod, toped up as needed. Intakes were recorded by staff at the end of each week at 0900 h by doing a weigh back of the feed offered that week and leftover in the trough. At this time, the trough would be cleaned, orts discarded, and entirely fresh feed offered unless the feed leftover was fresh from the evening before or that day, in which case they would add it back. As outlined in Table 2, from wk 1 to 8, calves were offered a calf starter pellet (Mapleview HE Calf Starter; 20% CP; Wallenstein Feed and Supply Ltd.; Table 1), and during wk 9, calves were offered a diet that consisted of half calf starter pellet and half calf grower ration (Mapleview Veal Starter 2.5:1; 18% CP; Wallenstein Feed and Supply Ltd.) and 4% chopped straw (The Straw Boss Inc.), referred to as combo feed (Table 1) to transition the calves onto the calf grower feed. In wk 10 and 11, calves were offered the calf grower ration with 4% chopped straw (Table 1). Both the calf starter and calf grower feeds contained monensin at 52 and 22.3 mg/kg, respectively.

Barn staff performed health assessments daily by walking through the barn and individually assessing each calf. Staff assigned fecal scores to all calves twice a day (0600 and 1600 h) from d 1 to 28. The scoring system was modified from McGuirk (2008): normal consistency feces received a score of 1, semi-formed or pasty feces received a score of 2, runny feces that spread easily received a score of 3, and liquid feces devoid of solid material received a score of 4. For treatment decision purposes (as outlined in Supplemental Table S1; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023) scores 3 and 4 were considered abnormal (neonatal calf diarrhea), while a score of 4 was considered severe diarrhea (Schinwald et al., 2022). As described in Supplemental Table S1, score 4 was treated upon the first day of occurrence, and score 3 was only treated (on first day of occurrence) when with blood. The fecal data were summarized into those categories for analyses. Specifically, the proportion of abnormal fecal scores was calculated for each calf by summing the number of fecal scores that were 3 or 4, dividing by the total number of scores (56), and multiplying by 100 to get a percent. The proportion of severe fecal scores (score or 4) was calculated for each calf using the same method except only summing the number of fecal scores that were 4. Staff assigned respiratory scores to all calves twice a day (also at 0600 and 1600 h) from d 1 to 77 using a scoring system modified from the UC Davis respiratory scoring system (Aly et al., 2020). Calves were assigned points per symptom of bovine respiratory disease (**BRD**) as follows: 2 points for each eye discharge, cough, rapid or difficult breathing, and rectal temperature >39.5°C, 4 points for nasal discharge, and 5 points for droopy ears. Rectal temperatures were only taken when the calf had a score of 4 based on the visible symptoms to determine if treatment was needed. All scores were recorded directly into a Microsoft Excel (Microsoft Corp.) spreadsheet. The respiratory data were summarized into 3 categories: the proportion of respiratory scores with no symptoms of

Table 1. Ingredient and chemical composition (mean ± SD) of the calf starter, combo feed,[1] calf grower feed,[1] and milk replacer fed to all calves

Composition	Calf starter feed[2]	Combo feed	Calf grower feed[3]	Milk replacer[4]
Ingredient, % in ration				
Calf starter pellet[5]	100	48	—	—
Calf grower pellet	—	13.7	27.5	—
Whole corn	—	33.8	67.5	—
Chopped straw[6]	—	4	4	—
Oil	—	0.5	1	—
Chemical composition[7]				
DM, % DM	88.9 ± 0.56	89.0	89.8	98.0 ± 0.37
CP, % DM	20.3 ± 0.32	18.5	18.1	24.6 ± 0.49
NDF, % DM	22.7 ± 1.68	20.4	15.7	—
ADF, % DM	14.4 ± 1.10	12.0	6.9	—
Crude fiber, % DM	—	—	—	0.4 ± 0.33
Ash, % DM	6.7 ± 0.24	7.5	5.7	7.2 ± 0.15
NFC, % DM	45.5 ± 1.68	49.6	54.8	—
Starch, % DM	28.9 ± 0.51	32.6	39.4	—
Fat,[8] % DM	4.1 ± 0.34	5.5	6.8	16.4 ± 0.70
Lactose,[9] % DM	—	—	—	51.9 ± 0.33
Ca, % DM	0.9 ± 0.10	0.9	1.2	1.0 ± 0.03
P, % DM	0.5 ± 0.02	0.4	0.5	0.8 ± 0.01
Na, % DM	0.4 ± 0.06	0.4	0.4	0.5 ± 0.04
ME,[10] Mcal/kg of DM	3.0 ± 0.04	3.1	3.3	4.6 ± 0.03

[1]Only 1 sample of each of the combo and calf grower feeds was available and analyzed.

[2]Calf starter pellet (Mapleview HE Calf Starter) was supplied by Wallenstein Feed and Supply Ltd., which included 52 mg/kg of monensin.

[3]The calf grower ration (Mapleview Veal Starter 2.5:1) in the calf-grower feed was supplied by Wallenstein Feed and Supply Ltd., which included 22.3 mg/kg of monensin. The ingredients included were whole corn, Mapleview Veal Starter Supplement (MON/CHL), and soya oil mix.

[4]Milk replacer was supplied by Mapleview Agri Ltd.

[5]Calf starter pellet included ingredients: corn ground-500, soybean meal, soya hulls, wheat Ontario ground-650, alfalfa meal, distillers dried grains with solubles–corn, lignin sulfonate (pellet binder), limestone B2, molasses liquid condensate, sugar, SoyPlus (Landus), tallow, salt feed mix (Wallenstein Feed and Supply Ltd.), dicalcium phosphate, micronutrient premix, vitamin ADE 8/1.25/8, magnesium oxide 56%, calf micronutrient premix, feed flavor, Selisseo 0.1% (Adisseo), cobalt 10,000 premix, zinc oxide 72%, and vitamin E 50% adsorbate.

[6]Chopped straw was supplied by The Straw Boss Inc.

[7]Chemical analysis was done by A&L Laboratory Services Inc. (London, ON, Canada).

[8]Fat for the rations is crude fat. Fat for the milk replacer was done by an acid hydrolysis test (AOAC International, 1995: Method 954.02).

[9]Lactose is assumed to be 100 − CP − fat − ash.

[10]Metabolizable energy was calculated using NRC (2001) equations.

Table 2. Feeding program for all calves

Period	Week	Trial days	Milk replacer (reconstituted with water; L/d)	Milk replacer (powder; g/d)	Ad libitum solid feed
Milk-fed	1	1–7	5	520	Calf starter
	2	8–14	5	650	Calf starter
	3	15–21	6	780	Calf starter
	4	22–28	7	910	Calf starter
	5	29–35	7	910	Calf starter
	6	36–42	6	780	Calf starter
	7	43–49	3	520	Calf starter
	8	50–56	2.5	325	Calf starter
Postweaning	9	57–63	—	—	Combo + straw (4%)
	10	64–70	—	—	Calf grower + straw (4%)
	11	71–77	—	—	Calf grower + straw (4%)

BRD, the proportion of respiratory scores ≥4, and the proportion of respiratory scores ≥5. Calves with a score ≥5 were diagnosed with BRD (McGuirk and Peek, 2014). Given the low proportion of respiratory scores ≥5, the proportion of respiratory scores ≥4 was also considered, as this grading has also been used to assess

BRD in calves (Love et al., 2014). Treatment of disease was determined based on the health scores and followed a treatment protocol, as outlined in Supplemental Table S1. Following the third treatment for diarrhea and fourth respiratory treatment, no further antimicrobial treatments were administered for diarrhea and BRD, respectively. All medications administered were recorded by barn staff. For each calf mortality, barn staff recorded the suspected cause of death.

Upon arrival to the facility, calves received an intranasal viral vaccine (Inforce-3; Zoetis Canada Inc.) and an antimicrobial medication (Draxxin; Zoetis Canada Inc.). At 7 d following arrival, calves received an intranasal bacterin (Once-PMH; Merck Animal Health), and at 14 and 28 d following arrival they received an injectable modified-live vaccine (Vista Once; Merck Animal Health).

Treatments Allocation

Calves were randomly allocated to 1 of 3 treatments, which included (1) receiving 3 g/d of dried (powder) EP extract split over 2 milk feedings from d 14 to 28 (**E14**; n = 80 calves), (2) receiving 3 g/d of EP split over 2 milk feedings from d 1 to 56 (**E56**; n = 80 calves), and (3) receiving no EP (control group; **CON**; n = 80 calves). For the E14 treatment, we were interested to see if a targeted supplementation of EP from d 14 to 28 following arrival would be as beneficial as supplementing it across the whole milk-fed period. Day 14 to 28 was chosen based on the estimated age of the calves at arrival to the facility (5–14 d of age) and the literature-based time when respiratory illness is first detected in preweaning dairy calves (Windeyer et al., 2014; Urie et al., 2018), thus to provide calves with EP while most susceptible to illness. For each room, before calf arrival, treatments were randomly allocated to pods using a random number generator in Microsoft Excel (Microsoft Corp.). This meant there was 25 or 30 calves per treatment in each room, and a total of 80 calves/treatment. The facility manager assigned the treatments to the pods to allow the primary author and fellow researchers to be blinded to the treatments.

At enrollment, following arrival to the facility, calves were weighed and randomly assigned to a stall number between 1 and 80, while ensuring the 3 treatment groups were kept within 0.45 kg of the average arrival BW. The treatments were identified by barn staff using 3 different colored ear tags (white = C, red = E14, and yellow = E56) to inform staff which treatment to feed each calf. The EP treatments were fed to the calves by the barn staff during milk feedings; the treatments (powdered EP) were mixed with 100 g of MR and added directly to the tank mix of milk. The staff did not know what the supplement was, nor the anticipated outcome, and had a busy schedule with multiple trials running, and therefore bias was not believed to be an issue for any of the data collected. The same number generator in Microsoft Excel (Microsoft Corp.) was used to select a subset of calves (13 calves per treatment in each room, 39 calves per treatment, total of 117 calves) for collection of additional measures including rectal temperature, blood haptoglobin, cytokine (IL-10, IL-6, and TNF-α) concentrations, and white blood cell (**WBC**) count and differentiation. Two to 3 calves were randomly selected from each pod to have them as equally distributed among the pods as possible.

Sample Size Calculation

Sample size calculations were conducted through a power analysis (Morris, 1999; Hintze, 2008) to ensure an adequate number of experimental units (calves) per treatment. It was estimated that a calf in the control group would have a preweaning ADG of 0.60 kg/d (SD = 0.2 kg/d; CV = 33%). With a 10% predicted difference, calves on the EP treatments would have an ADG of 0.66 kg/d (SD = 0.2 kg/d). Using 95% confidence interval (CI) and 80% power, a minimum sample size of 80 calves per treatment group or 240 calves total was determined to be required. It was estimated that the incidence of calf diarrhea and respiratory disease in the control group would be 20 and 30% (SD = 10%), based on historical facility data. We estimated that 80 calves per treatment group (n = 240 total), at 95% CI and 80% power, would allow us to detect a 15percentage-point difference in incidence of these health conditions.

Measurements and Sample Collection

Barn staff (a total of 6 individuals) collected all measurements, except blood samples, rectal temperatures, and hip heights, which were collected by the primary author with the assistance of fellow researchers (a total of 4 individuals). Of the 6 barn staff, only 2 did health scoring.

Blood sampling was completed on d 1, 14, 28, and 57 in the morning. On d 1, blood was collected from all calves into a 10-mL vacutainer red-top glass blood collection tube (BD), while an additional sample was collected from the subset calves into a 10-mL vacutainer lavender-top plastic blood collection tube with K2 EDTA (BD). On d 14, 28, and 57, the same 2 samples were collected from subset calves only. All blood samples were taken from the jugular vein using a 20 gauge × 2.54-cm vacuette multiple-use drawing needle (greiner BIO-ONE) and a vacutainer tube holder (BD). The sampler gently restrained the calf with their body

in the calf's stall for collection. Lavender-top tubes were gently inverted approximately 8 times immediately after collection to mix the blood with the K2 EDTA. Blood tubes were placed on ice in an insulated cooler after collection. Upon arrival at the University of Guelph, all lavender-top tubes were delivered to the Animal Health Laboratory (AHL; University of Guelph) where they were prepared and analyzed to determine the total WBC count and differentiation (segmented neutrophil, band neutrophil, lymphocyte, monocyte, eosinophil, and basophil counts). All red-top tubes, having had sufficient clotting time during the travel back to the University of Guelph, were placed in a centrifuge (Sorvall Legend RT) upon arrival and centrifuge cooling was completed. The samples were then centrifuged at 1,000 × *g* for 10 min at 4°C to obtain serum. Using disposable plastic pipets (Fisher Scientific), serum was aliquoted as follows: on d 1, a few drops of serum were placed on a hand-held digital refractometer (MISCO) to determine serum total protein (**STP**) and confirm passive immunity status, with failed transfer of passive immunity (**FTPI**) identified for STP values <5.2 g/dL (Renaud et al., 2020). The rest of the serum for the nonsubset calves was discarded, while for all 4 blood sampling days, the remaining serum for the subset calves was divided in triplicate (2 for analyses, 1 spare) into 0.5-mL snap-cap tubes (Fisher Scientific) and frozen at −20°C. Later, serum samples from each subset calf were delivered to the AHL (in an insulated container with freezer packs) where they were analyzed for haptoglobin content (g/L), using an in-house assay based on the work of Makimura and Suzuki (1982) and Skinner et al. (1991). At the end of the trial, serum samples were also sent to Eve Technologies Corp. (Calgary; in an insulated container with dry ice), where they were analyzed for cytokines. A multiplexing analysis was performed using the Luminex 200 system (Luminex). Three markers were simultaneously measured in the samples using a Custom Bovine Cytokine 3-Plex Magnetic Bead Assay (MilliporeSigma) according to the manufacturer's protocol. The 3-plex consisted of IL-6, IL-10, and TNF-α. Assay sensitivities of these markers range from 0.57 to 10.88 pg/mL for the 3-plex. The intra-assay %CV for all 3 cytokines was <10%, while the interassay %CV was <15% for IL-6 and TNF-α, and <10% for IL-10. Assessed blood parameters were chosen based on known EP effects reported in the literature to date (described above), as well as budget and laboratory constraints. No studies were identified that assessed acute phase proteins such as haptoglobin; we chose to assess haptoglobin in this study because EP has been reported to affect inflammation.

On d 1, 14, 28, and 57, during blood sampling, a rectal temperature (Accuflex 10 Flexible Digital Thermometer) was also taken and recorded on subset calves.

The MR refusals were determined by measuring any refused MR at each feeding and recorded for the purposes of monitoring health and calculating the total MR intake per calf at the end of the trial (by subtracting it from the total kg of MR offered). The solid feed refusals were determined at the end of each week by weighing the orts and recorded to later calculate weekly solid feed intakes per pod (by subtracting it from the total kg of feed offered that week). Samples of the MR and solid feed rations were taken biweekly. These samples were frozen at −20°C for later analysis. Samples were later thawed and individually placed in a drying oven at 60°C for 48 h to determine the DM content. Dry matter intake was calculated by multiplying the kg of solid feed consumed by the DM of the corresponding MR or solid feed ration sample for that week of data. Samples of solid feed were ground through a 1-mm sieve (Model 4 Wiley Laboratory Mill, Thomas Scientific) and were sent to A & L Canada Laboratories Inc. for chemical composition analyses. Calf starter, combo, and finisher feed samples were analyzed for ash (550°C; AOAC International, 2000: method 942.05), CP (N × 6.25; AOAC International, 2000: method 990.03; Leco FP-628 Nitrogen Analyzer), ADF (AOAC International, 2000: method 973.18), NDF with heat-stable α-amylase and sodium sulfite (AOAC International, 2000: method 2002.04), starch (heat-stable amylase and amyloglucosidase; AOAC International, 2000: method 996.11), and crude fat (AOAC International, 2000: method 920.39; Ankom XT15). Milk replacer samples were also analyzed at A & L Canada Laboratories Inc. for CP (N × 6.25; AOAC International, 2000: method 990.03; Leco FP-628 Nitrogen Analyzer), crude fiber (AOAC International, 2000: method Ba 6a-05; Ankom Bag Technology), and fat (total fat by acid hydrolysis using the ANKOM HCl Hydrolysis System; AOAC International, 1995: method 954.02 Section 4.5.02). Metabolizable energy was calculated for all samples using National Research Council (NRC, 2001) equations, and then the average ME was calculated for the MR and each of the solid feed rations. The weekly DMI values were then multiplied by the average ME of the corresponding MR or solid feed ration sample for that week of data to determine ME intake. For the milk-fed weeks, the ME intake from the MR and solid feed were summed to get a total. The total weekly ME intakes were then divided by pod gain to determine pod feed conversion rate (**FCR**) on a ME basis.

Table 3. Analysis results (mean ± SD) of the *Echinacea purpurea* samples fed to calves on those treatments that received it

Component	Result
DM (%)	97.3 ± 0.10
Aerobic plate count (cfu/g)	<10.0 ± 0.00
Phenolics (mg/kg)	
Chlorogenic acid	<60.0 ± 0.00
Cichoric acid	2,332.5 ± 59.10
Echinacoside	<60.0 ± 0.00
Caftaric acid	1,587.5 ± 41.13
Total phenolics	3,920.0 ± 100.00

Samples of the EP were taken at the beginning of each month of the trial and frozen at −20°C until the end of the trial. These samples were shipped to Eurofins Food Integrity and Innovation laboratory, where they were analyzed for percent moisture (United States Pharmacopeial Convention, 2019, modified) to determine DM content, aerobic plate count (standard plate count; cfu/g; FDA BAM Ch.3, AOAC 966.23, CMMEF Ch. 8, EML), and phenolic content including chlorogenic acid, cichoric acid, echinacoside, caftaric acid, and total phenolics (mg/kg; Sakakibara et al., 2003, Bauer and Wagner, 1991). Averages were calculated and reported in Table 3.

Calf BW was measured at 0900 h on arrival (d 0), and on d 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, and 77 using a calibrated platform scale (Tru-Test weigh platform; Tru-Test DATAMARS). Hip height was measured in the morning on d 1, 14, 28, and 57 of the trial.

Statistical Analyses

All statistical analyses were conducted using SAS 9.4 software (SAS Institute Inc., 2013), except diarrhea and BRD diagnoses and corresponding medication administrations, which were analyzed using Stata 16 (StataCorp LP). All data were imported into the statistical software programs from Microsoft Excel (Microsoft Corp.), where it was checked for completeness. In SAS, data were assessed for normality using the UNIVARIATE procedure. When parameters did not meet the assumptions of normality, a $\log_{10}$ transformation was used and successfully transformed the data to a normal distribution for the analysis. Transformed variables included monocytes, IL-6, IL-10, and TNF-α, abnormal and severe fecal scores, rectal temperature, electrolyte doses, and the proportion of respiratory scores ≥4 (milk-fed and postweaning periods). The proportion of respiratory scores ≥5 was not able to be normalized, and thus was categorized and presented as the risk of having BRD (1 = the calf had at least 1 respiratory score ≥5, 0 = the calf had no respiratory scores ≥5). Statistical differences were declared as significant at $P \leq 0.05$, and marginally significant at $0.05 < P \leq 0.10$.

Multivariable regression analyses were conducted for all parameters except medication administrations. The GLIMMIX procedure was used for all linear regression repeated measures analyses and logistic regression models, while the MIXED procedure was used for all linear models not containing repeated measures. For all repeated measures analyses with unequal time spacing, the covariance structure compound symmetry was used. For the repeated measures analyses with equal time spacing, the covariance structures compound symmetry, heterogeneous compound symmetry, heterogeneous first-order autoregressive, first-order autoregressive, and unstructured were tested and the one with the lowest BIC value was used for the analysis. The variables analyzed with logistic regression were the risk of BRD and mortality for both the milk-fed and postweaning periods. For each model, room was a block, kept constant as a fixed effect in each model to account for potential differences between the rooms at the facility. The covariates tested in each model were the categorical variables of FTPI (yes or no; 31.7% of calves had FTPI) and source (auction, drover 1, or drover 2), and the continuous variable arrival BW (average = 47.6 kg, minimum = 42.2 kg, maximum = 53.1 kg). The d 1 rectal temperatures were removed from the data and used as an additional covariate, kept constant as a fixed effect, for the rectal temperature analysis. Failed transfer of passive immunity and arrival BW were correlated, so if both were marginally significant or significant for a particular parameter, the more biologically related covariate was chosen to remain in the model. The baseline parameters were compared between treatments to ensure they did not vary. This was done using linear (arrival BW and STP) and logistic (FTPI) regression models and a chi-squared test (source). For any variables that were marginally significant or significant, the interaction of it with treatment was assessed. For the repeated measures analyses, the interaction of treatment and day was included. A manual backward stepwise process was used to eliminate variables that were not significant or marginally significant in the models, although treatment, day per week (for the repeated measures analyses), and room were kept in the models regardless of their outcome. The calf was the experimental unit in all analyses, except grain consumption and FCR, because grain consumption was measured at the pod level.

In Stata 16 (StataCorp LP), Cox proportional hazard models were created to access the effect of treatment on medication administration for diarrhea and BRD. A logistic regression model was used to determine the

Table 4. Multivariable linear regression model of the variables associated with blood haptoglobin concentration, segmented neutrophil count, lymphocyte count, and segmented neutrophils/lymphocytes ratio (N:L ratio)

	Haptoglobin (g/L)			Segmented neutrophils ($\times 10^9$/L)			Lymphocytes ($\times 10^9$/L)			N:L ratio		
Variable	Estimate	SE	*P*	Estimate	SE	*P*-value	Estimate	SE	*P*-value	Estimate	SE	*P*-value
Intercept	0.24	0.034	<0.001	3.93	0.324	<0.001	3.14	0.206	<0.001	1.29	0.092	<0.001
Treatment												
CON[1]	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
E14[2]	−0.05	0.026	0.07	−0.41	0.309	0.19	0.34	0.190	0.08	−0.16	0.087	0.06
E56[3]	−0.05	0.026	0.06	−0.57	0.311	0.07	0.38	0.190	0.05	−0.20	0.087	0.02
Day												
1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
14	−0.02	0.033	0.49	−1.15	0.258	<0.001	0.78	0.115	<0.001	−0.54	0.076	<0.001
28	0.01	0.033	0.84	0.10	0.258	0.70	0.98	0.115	<0.001	−0.28	0.076	<0.001
57	0.09	0.034	0.01	−0.33	0.260	0.20	1.56	0.116	<0.001	−0.47	0.077	<0.001
Room												
1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
2	0.04	0.026	0.12	0.23	0.310	0.46	0.25	0.191	0.20	0.02	0.087	0.82
3	0.03	0.027	0.29	0.68	0.310	0.03	0.15	0.191	0.42	0.18	0.087	0.04
Source												
Auction	Referent	—	—	—	—	—	Referent	—	—	—	—	—
Drover 1	−0.06	0.029	0.03	—	—	—	−0.43	0.211	0.04	—	—	—
Drover 2	−0.03	0.024	0.23	—	—	—	0.05	0.175	0.76	—	—	—

[1]CON = control, calves (n = 39) received no *Echinacea purpurea*.
[2]E14 = calves (n = 39) received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 14–28.
[3]E56 = calves (n = 39) received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 1–56.

effect of treatment on recurrent medication administrations. Note that data from both the milk-fed and post-weaning periods were included in determining the risk of receiving the respiratory treatments. Similar model building strategies were used, as outlined above, and model fit was determined by evaluating the assumption of proportional hazards in the hazard models.

RESULTS

The baseline parameters did not differ between treatment groups, including arrival BW ($P = 0.76$), STP ($P = 0.34$), FTPI ($P = 0.91$), and source ($P = 0.94$).

Blood

Calves in the E14 and E56 treatments had marginally significantly lower levels of haptoglobin than CON calves (Table 4). Calves in the E56 treatment had marginally significantly less segmented neutrophils than CON calves (Table 4). Calves in the E14 treatment had marginally significantly more lymphocytes, while E56 calves had more lymphocytes than CON calves (Table 4). Calves in the E14 treatment had a marginally significantly lower segmented neutrophil/lymphocyte ratio (**N:L**) ratio, while E56 calves had a lower N:L ratio than CON calves (Table 4). There was a treatment × arrival BW interaction for eosinophils ($P = 0.005$; Figure 1). Calves in the CON treatment had increased eosinophil counts with increased arrival BW, whereas calves in the E14 and E56 treatments had consistent eosinophil counts regardless of arrival BW.

We observed no detected effect of the EP treatments on total WBC, band neutrophil, monocyte, and basophil (Supplemental Table S2; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023)) counts, nor IL-10, 1L-6, and TNF-α levels (Supplemental Table S3; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023).

Health

A marginally significant treatment × FTPI × d interaction ($P = 0.09$) was detected for rectal temperature; on d 14, of the calves with FTPI, E14 calves had a marginally significantly lower temperature than CON calves ($P = 0.10$; Figure 2). Additionally, on d 28, of calves without FTPI, E14 calves had a higher temperature than CON calves ($P = 0.02$), and E56 calves had a marginally significantly higher temperature than CON calves ($P = 0.08$; Figure 2).

In the milk-fed period, there was a treatment × source interaction ($P = 0.009$) for the proportion of respiratory scores with no symptoms of bovine respiratory disease; of calves sourced from auction, E14 calves had a marginally significantly higher proportion of

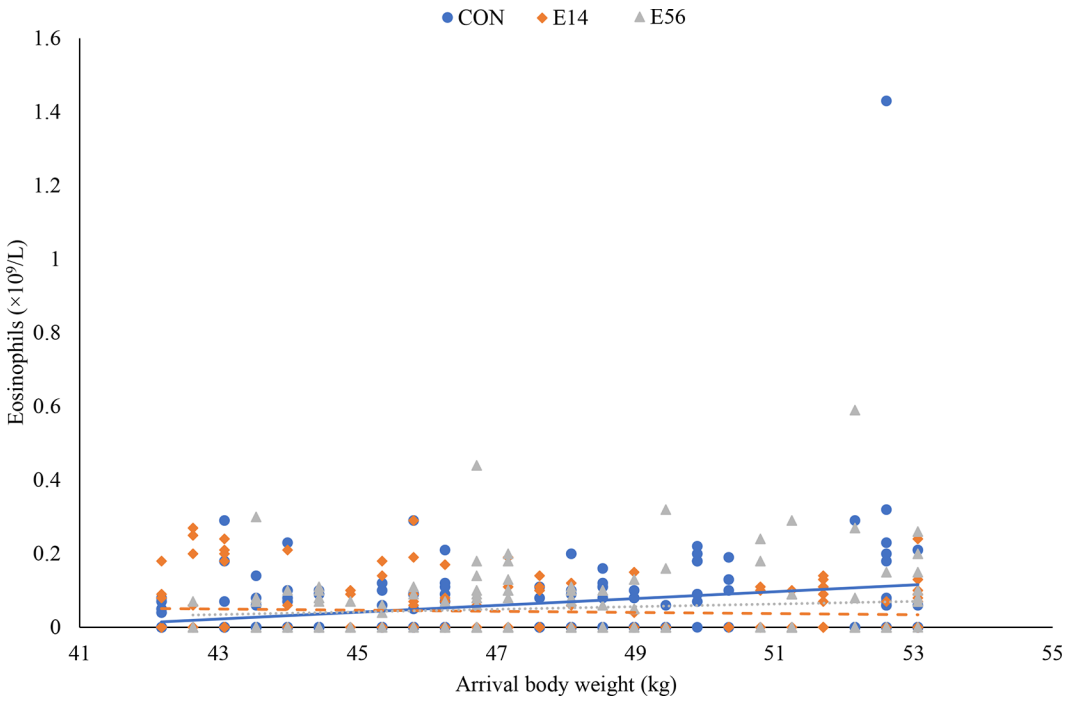


Figure 1. Eosinophils by treatment and arrival BW. Each dot represents a calf. The treatments included CON = control (calves received no *Echinacea purpurea*), E14 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 14–28, and E56 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 1–56. n = 39 calves/treatment.

scores with no symptoms of BRD than CON calves ($P = 0.08$) and E56 calves had a higher proportion of scores with no symptoms of BRD than CON calves ($P = 0.03$; Figure 3). Conversely, of calves sourced from drover 2, E14 calves had a marginally significantly lower proportion of scores with no symptoms of BRD than CON calves ($P = 0.08$) and E56 calves had a lower proportion of scores with no symptoms of BRD than CON calves ($P = 0.01$; Figure 3). In the milk-fed period, we observed no detected effect of the EP treatments on the proportion of abnormal nor severe fecal scores, electrolyte doses (Supplemental Table S4; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023), the risk of receiving diarrhea treatment (Supplemental Table S5; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023), the proportion of respiratory scores ≥4 (Supplemental Table S4), the risk of BRD, and the risk of mortality (Supplemental Table S6; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023).

In the postweaning period there was a treatment × source interaction ($P = 0.01$) for the proportion of respiratory scores ≥4; of calves sourced from auction, E14 calves had a marginally significantly lower proportion of respiratory scores ≥4 than CON calves ($P = 0.06$; Figure 4). Of calves sourced from drover 1, E14 calves had a higher proportion of respiratory scores ≥4 than CON calves ($P = 0.02$) and had a marginally significantly higher proportion of respiratory scores ≥4 than E56 calves ($P = 0.10$; Figure 4). Of calves sourced from drover 2, E14 calves had a higher proportion of respiratory scores ≥4 than CON calves ($P = 0.05$; Figure 4). In the postweaning period, we detected no effect of the EP treatments detected on the proportion of scores with no symptoms of BRD (Supplemental Table S9; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023), the risk of BRD, the risk of mortality (Supplemental Table S10; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023), and the risk of receiving respiratory treatment (Supplemental Table S11; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023). Throughout the whole trial, a total of 22 calves (9.17%) died.

Intake and Growth

In the milk-fed period, we detected no effect of the EP treatments detected on any of the intake nor growth parameters, including MR intake, grain intake, FCR (Supplemental Table S7; https://doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023), hip height, weekly BW, and ADG (Supplemental Table S8; https://

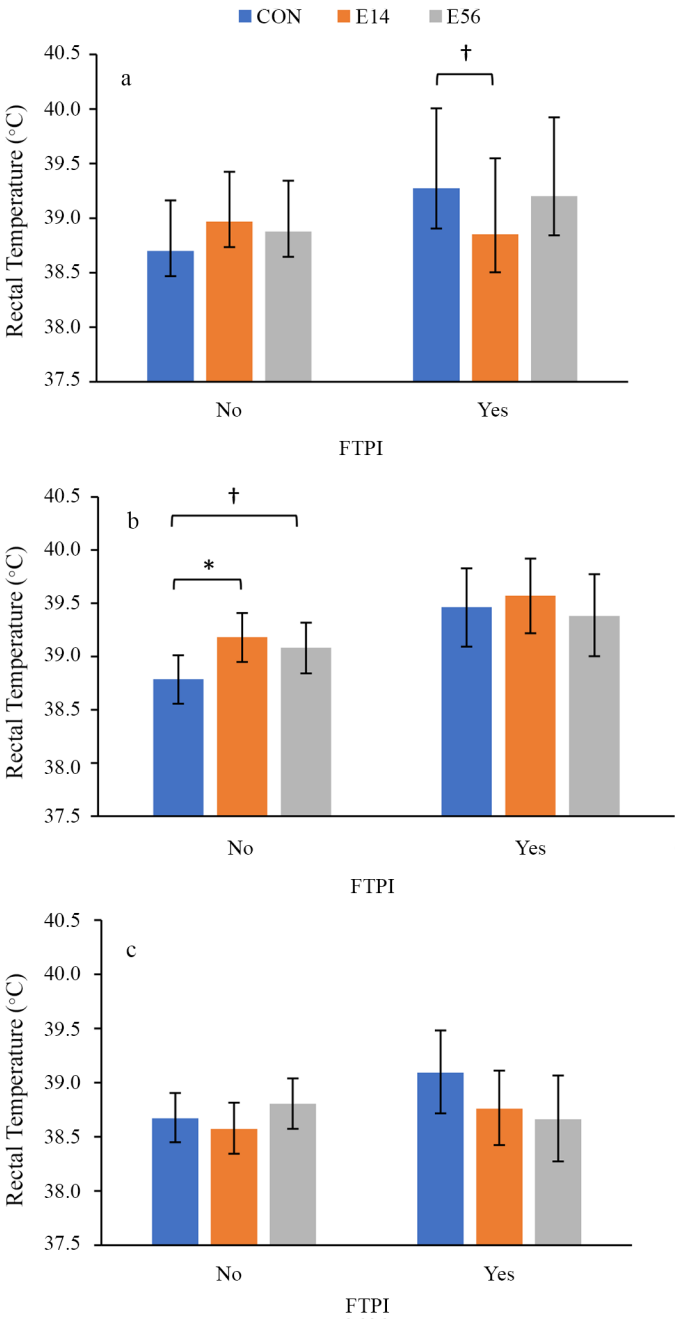


Figure 2. Rectal temperature (back-transformed mean ± 95% CI) by day (a = 14, b = 28, and c = 57), treatment and failed transfer of passive immunity (FTPI) status, defined as serum total protein (STP) values <5.2 g/dL. Within FTPI status (no or yes), significant differences ($P \leq 0.05$) detected between treatments are denoted by *, and marginally significant differences ($0.05 < P \leq 0.1$) are denoted by †. The treatments included CON = control (calves received no *Echinacea purpurea*), E14 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 14–28, and E56 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 1–56. n = 39 calves/treatment.

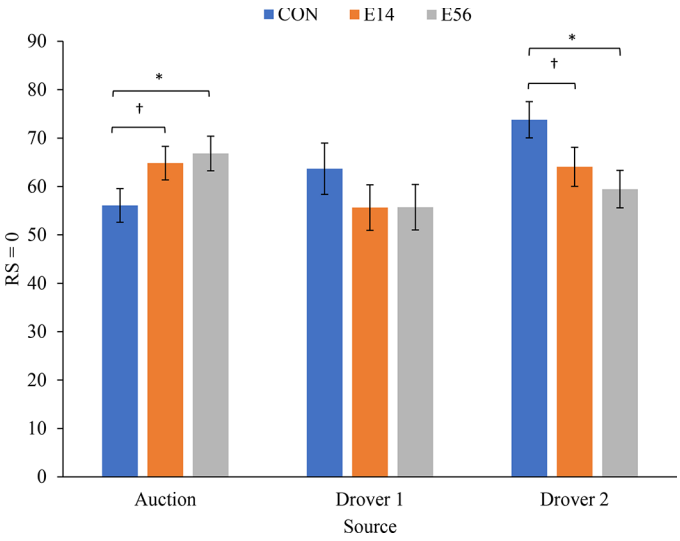


Figure 3. Proportion (mean ± SE) of respiratory scores with no symptoms of bovine respiratory disease (RS = 0) in the milk-fed period by treatment and source. Within each source, significant differences ($P \leq 0.05$) detected between treatments are denoted by *, and marginally significant differences ($0.05 < P \leq 0.1$) are denoted by †. The treatments included CON = control (calves received no *Echinacea purpurea*), E14 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 14–28, and E56 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 1–56. n = 80 calves/treatment.

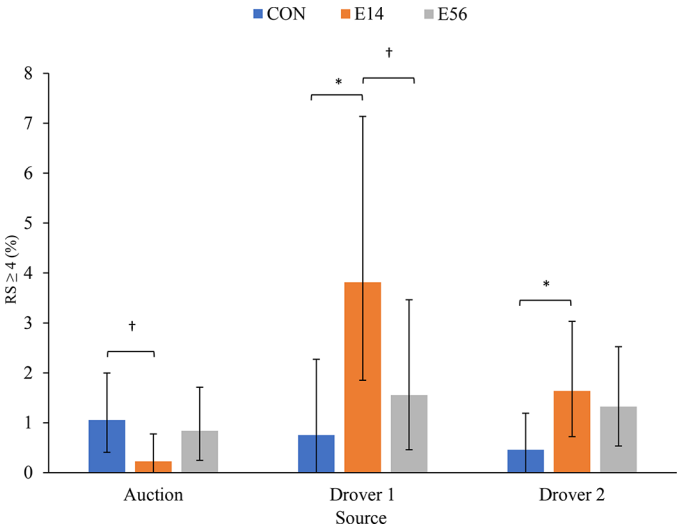


Figure 4. Proportion (back-transformed mean ± 95% CI) of respiratory scores ≥4 (RS ≥4) in the postweaning period by treatment and source. Within each source, significant differences ($P \leq 0.05$) detected between treatments are denoted by *, and marginally significant differences ($0.05 < P \leq 0.1$) are denoted by †. The treatments included CON = control (calves received no *Echinacea purpurea*), E14 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 14–28, and E56 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 1–56. n = 80 calves/treatment.

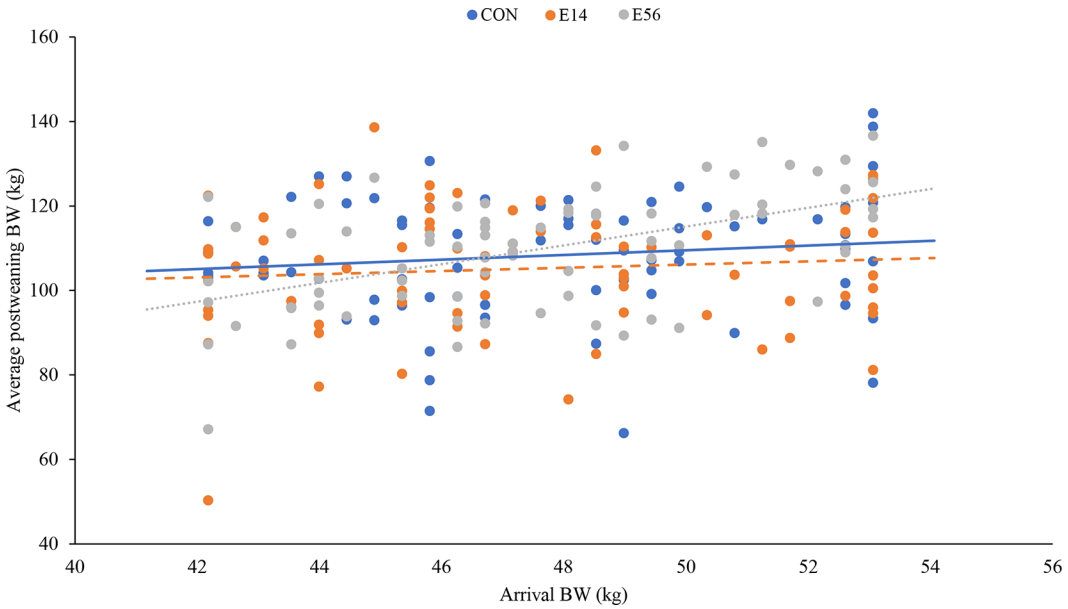


Figure 5. Average postweaning period weekly BW by treatment and arrival BW. Each dot represents a calf. The treatments included CON = control (calves received no *Echinacea purpurea*), E14 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 14–28, and E56 = calves received 3 g/d of *Echinacea purpurea* split over 2 milk feedings from d 1–56. n = 80 calves/treatment.

doi.org/10.5683/SP3/ES2TLC; McNeil and DeVries, 2023). In the postweaning period, there was a treatment × arrival BW interaction ($P = 0.004$) for postweaning period weekly BW; of calves with heavier arrival BW, E56 calves had higher average postweaning weekly BW than CON calves ($P = 0.006$; Figure 5). We observed no detected effect of the EP treatments on grain intake, ADG, and FCR (Supplemental Table S9; https://doi.org/10.5683/SP3/ES2TLC).

DISCUSSION

This study is a thorough investigation of the effects of supplementing EP to dairy calves. *Echinacea purpurea* supplementation was associated with reduced haptoglobin levels, reduced segmented neutrophil counts and a lower N:L ratio, as well as increased lymphocyte counts. Additionally, EP supplementation was associated with a higher average postweaning body weight among calves with heavier arrival BW. Effects were detected to a greater extent in E56 rather than E14 calves. Supplements are a growing area of research, and rightfully so, as they have the potential to support dairy calves in various ways, including immunity, health, growth, and welfare, and can be easily implementable on farm. Calf-hood success is vital to the future productivity and sustainability of the dairy industry.

It was hypothesized that EP supplementation, especially for the whole milk-fed period (E56), would be associated with reduced inflammation (reduced haptoglobin) and stimulated immunity (increased cytokines and WBC counts) due to EP's active components, which would in turn minimize illness. In support of this, haptoglobin was marginally significantly lower in the E14 and E56 calves, indicative of lower inflammation (Murray et al., 2014). To our knowledge, no studies have assessed haptoglobin in response to only EP supplementation. In support of our finding, Fararh et al. (2017) reported that diseased calves supplemented with Tulathromycin and an essential oil mixture, which included EP, had lower haptoglobin levels than diseased calves treated with just Tulathromycin alone. Given the combination of supplements provided in that study, one cannot conclude their results are related solely to the EP. For this reason, in the following discussion of results from other EP supplementation studies, only those results from supplementation of EP alone are presented, unless otherwise stated (i.e., EP was supplemented in combination with other products).

Marginally fewer segmented neutrophils in the E56 calves would be indicative of lesser infection or inflammation (Doherty et al., 2007; Cuevas-Gómez et al., 2020). This is further evidence (in addition to reduced haptoglobin) of reduced inflammation in E56 calves. Although there are conflicting results in the literature regarding the effect of EP supplementation on neutrophil counts, including decreased levels in rabbits (Ahmed et al., 2008) and chickens (Nosrati et al., 2017), increased levels in chickens (Dehkordi et al., 2011; Enany et al., 2017), and no detected effect in

pigs (Maass et al., 2005), dogs (Torkan et al., 2015), and calves (Ayrle et al., 2021). Greater lymphocytes were detected in calves supplemented with EP, especially those receiving it throughout the whole milk-fed period, indicative of an immune system more prepared to defend the body against pathogens (Orakpoghenor et al., 2019), supporting our hypothesis of stimulated immunity. Similarly, EP supplementation for a longer duration, compared with a shorter duration and control, was associated with higher lymphocyte count in chickens (Dehkordi et al., 2011). Likewise, supplementation of EP has been associated with increased lymphocyte counts in dogs (Torkan et al., 2015) and chickens (Enany et al., 2017; Nosrati et al., 2017). Conversely, some researchers did not detect any effect of EP supplementation on lymphocyte counts in pigs (Maass et al., 2005), chickens (Gharieb and Youssef, 2014), and calves (Ayrle et al., 2021). The N:L ratio has been used as a measure of illness and stress in humans and animals. In fact, it was reported to be higher in humans with advanced or aggressive cancer (Guthrie et al., 2013) and severe cases of COVID-19 (Zheng et al., 2020), and in sick or presumably stressed calves (Doherty et al., 2007; Cuevas-Gómez et al., 2020). A lower N:L ratio was detected in calves supplemented with EP, especially those receiving it throughout the whole milk-fed period, indicative of less illness or stress. Similarly, Nosrati et al. (2017) detected a lower N:L ratio in chickens supplemented with EP. Although, other researchers detected no effect of EP supplementation on the N:L ratio including in chickens (Gharieb and Youssef, 2014; Jahanian et al., 2017) and calves (Ayrle et al., 2021). Overall, the lower haptoglobin, segmented neutrophils, and N:L ratio, as well as higher lymphocytes associated with the treatments receiving EP supplementation in the present study are likely due to the active components in the EP, namely the phenolics, alkamides, and polysaccharides, causing antioxidant, anti-inflammatory, and immune stimulatory effects, respectively. Comparing the results of this study to Ayrle et al. (2021), the EP supplemented in our study had higher concentrations of phenolics. For example, considering cichoric acid—the primary phenolic in EP (Dalby-Brown et al., 2005)—the EP in our study contained 2,332.5 ± 59.10 mg/kg, whereas Ayrle et al. (2021) reported the EP they supplemented only contained 313.8 mg/kg. Cichoric acid in the diet produces antioxidant activity, which supports health and a properly functioning immune system (Puertollano et al., 2011). Therefore, the phenolic content difference is likely a major reason behind the differences detected. We were unable to locate the resources to determine the alkamide and polysaccharide content in the EP supplemented, which was also not reported by Ayrle et al. (2021) for their EP, although those components were likely to vary as well, which would further affect the differences in results.

Various blood parameters did not, however, align with our hypothesis. Of calves with heavier arrival BW, E14 and E56 calves' eosinophil counts remained relatively consistent, whereas CON calves' eosinophil counts were elevated. It is possible that a heavier arrival BW in addition to EP supplementation was favorable for health, preventing an increase in eosinophils. However, Ayrle et al. (2021) detected lower eosinophil counts in calves receiving a low EP dose compared with control, but no difference for calves receiving a high dose. Conversely, Ahmed et al. (2008) reported that rabbits supplemented with a high dose of EP had increased eosinophils compared with lower doses and their control. In general, most researchers have reported no effects of EP supplementation on eosinophil count, including in pigs (Maass et al., 2005) and chickens (Dehkordi et al., 2011; Gharieb and Youssef, 2014).

Despite the expectation that WBC, band neutrophil, basophil, and monocyte counts would all be lower in E14 and E56 calves, no differences were detected. The increase in lymphocytes and decrease in neutrophils associated with EP supplementation in the present study may have eliminated a change in WBC count. Similarly, other researchers did not detect an effect of EP supplementation on WBC count in pigs (Maass et al., 2005), chickens (Gharieb and Youssef, 2014), or calves (Ayrle et al., 2021). However, supplementation of EP has been associated with an increased WBC count in various studies, including in chickens (Dehkordi et al., 2011; Enany et al., 2017) and dogs (Torkan et al., 2015). In those studies, such an increase may be explained by the EP activating the immune system (as illness would), causing an increase in WBC count to prepare the body to fight infection. To our knowledge there is a lack of data available in the literature on the effect of EP supplementation on band neutrophils and basophil counts. Similar to our results, Torkan et al. (2015) also did not detect any effect of EP supplementation on band neutrophil counts in dogs, and Maass et al. (2005) and Torkan et al. (2015) also did not detect any effect of EP supplementation on basophil counts in pigs and dogs, respectively. Alternatively, Ahmed et al. (2008) reported decreased basophil counts in rabbits. Similar to our results, other researchers did not identify any effects of EP supplementation on monocyte count in chickens (Gharieb and Youssef, 2014; Enany et al., 2017) and calves (Ayrle et al., 2021). Greater and decreased monocyte counts, however, were reported in chickens (Jahanian et al., 2017) and rabbits (Ahmed et

al., 2008), respectively. Eosinophils, band neutrophils, and basophils are all found in low quantities in calves (Knowles et al., 2000), which may explain the variable or lack of detected effects on these measures.

Contrary to our hypothesis, especially given the noted effects on haptoglobin and the N:L ratio, IL-10, IL-6, and TNF-α levels, which are known to regulate inflammation, did not vary with EP treatment. A likely explanation is that the EP supplemented had a low aerobic plate count ($<10.0 \pm 0.00$ cfu/g), as it is the bacterial endophytes that are reported in the literature to have the greatest effect on cytokines, as noted above. In general, IL-10 levels have not been well investigated in calves, whereas IL-6 and TNF-α have been more thoroughly investigated in calves, with higher levels associated with illness in several studies, including pneumonia (Akgul et al., 2019) and neonatal calf diarrhea (Beheshtipour and Raeeszadeh, 2020). Considering EP supplementation studies, cytokine results are variable. Schwarz et al. (2002) and Ayrle et al. (2021) did not detect any effect of EP supplementation on TNF-α levels in humans and TNF-α mRNA in calves, respectively. However, a low dose of EP supplementation was associated with lower IL-6 levels and a high dose of EP supplementation was associated with lower TNF-α levels in rodents (Yu et al., 2013). Conversely, in vitro human macrophages cultured in EP produced higher levels of IL-10, IL-6, and TNF- α than human macrophages cultured in a control medium (Burger et al., 1997). Likewise, Seckin et al. (2018) detected greater upregulation of TNF-α gene expression in calves receiving EP and *Pelargonium sidoides*. Due to the high variability in results, more research is needed to elucidate the effect of EP supplementation on cytokine levels. It is noteworthy that the majority of studies assessing the effect of EP supplementation were focused on blood markers and performance (intake and growth) measures but did not report on animal health outcomes.

Some, although not all, results aligned with the hypothesis that EP supplementation would improve calf health. Of those calves with FTPI, E14 calves had a marginally significantly lower temperature on d 14 than CON calves. The calves received the EP at 0600 h, while the temperatures were taken throughout the morning depending on the calves' location in the room. It is possible that the introduction of EP resulted in a lower rectal temperature in these immune compromised calves due to its anti-inflammatory properties. On d 28, of calves without FTPI, EP supplemented calves had a higher rectal temperature, especially those receiving it for just 2 wk. This was unexpected, because a high rectal temperature is a well understood symptom of illness (Hart, 1988), and therefore was expected to be lower in calves receiving EP, especially as the trial proceeded to give the EP time to take effect. It is possible the EP can cause a higher temperature by stimulating the immune system. Ayrle et al. (2021) reported that EP resulted in a higher rectal temperature in calves on low and high EP supplementation treatments compared with control, although on different days throughout the trial. Evidently more investigation is needed.

Contrary to our expectation, EP supplementation did not result in any improvement in fecal scores or reduced diarrhea treatment. Ayrle et al. (2021) reported a low dose of EP supplementation reduced the number of days with diarrhea in calves by 44%, although a high dose had no effect. The effects of EP on digestive health are lacking in the literature and require more investigation.

In the milk-fed period, of calves sourced from auction, EP supplementation was associated with a higher proportion of scores with no symptoms of BRD, especially in calves receiving it throughout the whole milk-fed period. The blood parameters indicative of improved immunity associated with the E14 and E56 treatments likely contributed to this result. It could be assumed that calves sourced from auction incurred more stress than those sourced from the drovers, and research underlines that stress can negatively affect the immune system (Hulbert and Moisá, 2016). Perhaps the effects of EP on health can be more evident in immune compromised or stressed animals. Surprisingly, in the milk-fed period, of calves sourced from drover 2, EP supplementation was associated with a lower proportion of scores with no symptoms of BRD, especially in calves receiving it throughout the whole milk-fed period. This was an unexpected finding, as the opposite result was reported in auction calves (as discussed above), yet no differences in blood markers, health, intake, or growth parameters were detected between calves sourced from drover 2 and auction. Despite preweaning differences, in the postweaning period, we detected no effect of the EP treatments detected on the proportion of scores with no symptoms of BRD.

Even though no differences were detected in the milk-fed period, in the postweaning period, of calves sourced from auction, E14 calves had marginally significantly better respiratory health (i.e., a lower proportion of respiratory scores ≥ 4) than CON calves, whereas, of calves sourced from drover 2, E14 calves had worse respiratory health than CON calves. Again, because we observed no differences detected in blood markers, health, intake, or growth parameters between calves sourced from drover 2 and auction, it is unknown why the opposite result occurred in auction versus drover 2 calves. Of calves sourced from drover 1, E14 calves had the worst respiratory health. Again, worse respiratory health in E14 calves compared with CON calves was

not expected. Despite this, a higher proportion of respiratory scores ≥4 in calves sourced from drover 1 is not surprising—for as compared with auction calves, drover 1 calves had more diarrhea, a lower lymphocyte count, higher IL-6 levels, less MR intake, and a 4× greater risk of mortality. Similarly, Ayrle et al. (2021) reported a low dose of EP supplementation to calves resulted in an increased number of days with respiratory disease during a follow-up period, although a high dose did not vary from control. Possibly, a shorter duration or lesser amount of EP supplementation can still stimulate the immune system in seemingly beneficial ways, as shown by some of the blood results in the present study, although is not enough to combat illness. Furthermore, such EP supplementation combined with immune system activation from disease may overload the calf's immune system (Aucoin et al., 2020), negatively affecting their ability to fight disease.

Echinacea supplementation had no detected effect in either the milk-fed or postweaning periods on the risk of BRD or risk of receiving respiratory treatment. This does not align with the blood parameters indicative of improved immunity, although is not surprising due to the conflicting respiratory score results. The minimal respiratory health benefits detected were unexpected because there has been a lot of research focused on EP as a supplement to support respiratory health. Research in humans is largely focused on the use of EP as a treatment rather than a preventative supplement. Taylor et al. (2003) reported treatment with EP did not improve symptoms of upper respiratory tract infections in children, whereas Weber et al. (2005) reported EP may reduce the risk of subsequent upper respiratory tract infections in children. Likewise, Aucoin et al. (2020) reported it may improve symptoms of acute respiratory infections and the common cold, especially when taken at the first sign of infection. Despite inconclusive results, there is a large consensus that EP consumption may have respiratory benefits; in fact, the COVID-19 pandemic brought even more attention to this herb through investigations on its efficacy in the treatment of COVID-19 infections (Aucoin et al., 2020). Additionally, a review by Ayrle et al. (2016) concluded that EP was 1 of 3 medicinal plants (out of 30 evaluated) with the most promise for the prevention or treatment of respiratory diseases in calves and piglets. Overall, the effects of EP supplementation on respiratory health, especially in calves, is lacking in the literature and requires more investigation, ideally including lung ultrasound to truly identify BRD status.

Contrary to expectation, EP supplementation had no detected effect on the risk of mortality in either the milk-fed or postweaning periods. To our knowledge, there is no other work on EP supplementation on mortality rates in calves for comparison. That said, Enany et al. (2017) reported that during an induced *Escherichia coli* infection in chickens, EP reduced morbidity and mortality rates, and Ahmed et al. (2008) reported lower mortality in rabbits supplemented with EP.

It was hypothesized that EP supplementation, through improving immunity and health, would allow more energy to be devoted to growth and, therefore, result in improved performance parameters. Only one such effect was detected: of calves with heavier arrival BW, those receiving EP throughout the whole milk-fed period had a higher postweaning weekly BW. It is sensible that calves with higher arrival BW would be predisposed to having larger BW long-term. Additionally, E56 calves had the lowest neutrophil count and N:L ratio and greatest lymphocyte count supporting inflammation reduction, immunity stimulation, and possibly improved affective state, all of which would likely support further growth postweaning.

Intake and growth in animals in response to EP supplementation has been thoroughly investigated in the literature. Similar to the large majority of our results, in many studies there was no effect of EP supplementation detected on performance parameters, including feed intake in rabbits (Ahmed et al., 2008) and chickens (Gurbuz et al., 2010; Dehkordi et al., 2011), BW gain in pigs (Maass et al., 2005) and chickens (Dehkordi et al., 2011), and FCR in piglets (Maass et al., 2005) and chickens (Jahanian et al., 2017). Although in some studies there were reported improvements in performance parameters associated with EP supplementation, including increased feed intake in *E. coli*-challenged chickens (Gharieb and Youssef, 2014), BW gain in chickens (Gharieb and Youssef, 2014; Enany et al., 2017; Nosrati et al., 2017), and lower FCR in pigs (Maass et al., 2005), rabbits (Ahmed et al., 2008), and chickens (Dehkordi et al., 2011). It is presumed that the active components in the EP supported growth, as hypothesized in the present study. Although only noted in a few studies, there were worse outcomes in performance parameters associated with EP supplementation, Gurbuz et al. (2010) reported lower ADG, and higher FCR in chickens supplemented with EP. There are many reasons for the variability in results across studies, including (1) the EP product used varied: handling factors such as growing medium, age of plant at harvest, part of plant used (root, aerial, or whole), and method of extraction affect the active component profile of the EP (Dehkordi et al., 2011; Haron et al., 2019); (2) the dose amounts and duration used varied, and (3) housing conditions and animal care varied, which are factors known to affect immunity and health. Additionally, there is minimal calf research on the effects of EP supplementation. Most research was conducted

with chickens, which are less-ideal comparisons. Of the calf studies available, and in many of the studies with other animal species, EP has been often supplemented in addition to other products; as challenging as it is to demonstrate the activity of a single feed additive, it is even more complex when a combination is used.

This study had a few notable limitations. First, health history and care (birthing management, vaccinations, colostrum and milk or grain feeding, housing, and so on) of the calves before arrival to the calf rearing facility for enrollment in the study was unknown and likely variable depending on the farm they came from. Also, the exact age of each calf was unknown; calf immunity and health can vary depending on age, therefore variable ages among calves could affect the results. Additionally, the details listed in the previous paragraph for the EP product used were unknown. We were unable to follow the calves past 77 d to see if there were long-term effects. Due to time and resource limitations, we were unable to enroll a larger number of calves to investigate more treatment regimens (dosage of EP/d and duration of supplementation) to identify the most effective. Due to facility design, calves were fed solid feed by pods rather than individually; the latter method would have been superior to be able to report solid feed results (intake and FCR) at the calf level. Finally, EP was only supplemented in the MR during the milk-fed period and was unable to be supplemented at the calf level postweaning due to group housing, although it would have been interesting to see if longer term supplementation through top dressing it on calf starter postweaning had any effects.

CONCLUSIONS

Supplementation of EP was associated with blood markers indicative of reduced inflammation and a stimulated immune system, particularly when fed throughout the whole milk feeding period, although minimal beneficial effects were detected on health and performance. Importantly, any negative effects identified were covariate dependent and, therefore, require more investigation. More research on EP supplementation to dairy calves is recommended to confirm and support its effects and identify the best EP product, dose and duration of supplementation to maximize its benefits.

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